

$$M\ddot{u}_n = k(u_n - u_{n-1}) - Q(u_n - v_{n-1})$$

$$M\ddot{v}_n = Q(u_{n+1} - v_n) - k(v_n - u_n)$$

Nastavek:

$$u_n = u_0 e^{i(kna - \omega t)}$$

$$v_n = v_0 e^{i(kna - \omega t)}$$

$$-\omega^2 M u_0 e^{i(kna - \omega t)} = k[u_0 e^{i(kna - \omega t)} - u_0 e^{i(k(n-1)a - \omega t)}] - Q[u_0 e^{i(kna - \omega t)} - v_0 e^{i(k(n-1)a - \omega t)}]$$

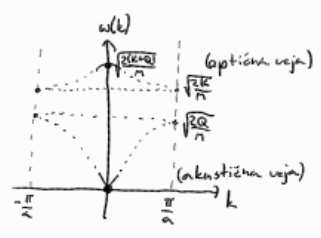
$$-\omega^2 M u_0 = k(u_0 - u_0 e^{-ika}) - Q(u_0 - v_0 e^{-ika})$$

$$-\omega^2 M v_0 = Q(u_0 e^{ika} - v_0) - k(v_0 - u_0)$$

$$\begin{bmatrix} k + Q - M\omega^2 & -k - Qe^{-ika} \\ -k - Qe^{ika} & k + Q - M\omega^2 \end{bmatrix} \begin{bmatrix} u_0 \\ v_0 \end{bmatrix} = 0$$

$$\omega^2 = \frac{k + Q}{M} \left[1 \pm \sqrt{1 - \frac{4kQ}{(k+Q)^2} \sin^2 \frac{ka}{2}} \right]$$

$$= \frac{k + Q}{M} \left[1 \pm \left(1 - \frac{4kQ}{(k+Q)^2} \left(\frac{ka}{2} \right)^2 \right) \right]$$



Lastni vektorji

- 1) $k=0$
- 2) $\omega=0$

$$\omega = \sqrt{\frac{2(k+Q)}{M}}$$

$$\begin{bmatrix} k + Q & -k - Q \\ -k - Q & k + Q \end{bmatrix} \begin{bmatrix} u_0 \\ v_0 \end{bmatrix} = 0$$

$$\begin{bmatrix} \dots \end{bmatrix} \begin{bmatrix} u_0 \\ v_0 \end{bmatrix} = 0$$

$$\rightarrow \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} u_0 \\ v_0 \end{bmatrix} \quad u_n = u_0 \quad v_n = u_0$$

$$\begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} u_0 \\ v_0 \end{bmatrix} \quad u_n = u_0 e^{-i\omega t} \quad v_n = -u_0 e^{-i\omega t} = u_0 e^{-i(\omega t - \pi)}$$

3) $k = \frac{\pi}{a}$

$$1) \omega = \sqrt{\frac{2Q}{M}}$$

$$\omega = \sqrt{\frac{2Q}{M}}$$

$$\begin{bmatrix} u_0 \\ v_0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad u_n = u_0 e^{i(n\pi - \omega t)} \quad v_n = u_0 e^{i(n\pi - \omega t)} \quad e^{in\pi} = (-1)^n$$

$$\begin{bmatrix} u_0 \\ v_0 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

3D nihanje (iz ravnine)



a_0 - dolžina neraztegnjene vzmeti
 $a > a_0$



$$F = k(\sqrt{a_0^2 + (u_2 - u_1)^2} - a_0)$$

$$\tan \alpha = \frac{u_2 - u_1}{a_0} \approx \sin \alpha \approx \alpha$$

ravnina table (skica od strani)

$$F_{\perp} = F \sin \alpha = k(\sqrt{a_0^2 + (u_2 - u_1)^2} - a_0) \frac{u_2 - u_1}{a_0} \approx k(1 - \frac{a_0}{a})(u_2 - u_1) = \tilde{k}(u_2 - u_1)$$

$$F_{\parallel} = F \cos \alpha = k(\sqrt{a_0^2 + (u_2 - u_1)^2} - a_0) \left(1 - \frac{(u_2 - u_1)^2}{a_0^2} \right) \approx k(a - a_0) + o((u_2 - u_1)^2) \approx 0 \text{ (med sosedi izničijo)}$$

(harmonski približek)

$$M\ddot{\vec{u}}_{\vec{R}} = \sum_{\vec{R}' \text{ sosed.}} \vec{F} = -\tilde{k} \sum_{\vec{R}' \text{ sosed.}} (u_{\vec{R}} - u_{\vec{R}}') \quad u_{\vec{R}} = u_0 e^{i\vec{k} \cdot \vec{R} - \omega t}$$

$$-\omega^2 M u_0 e^{i\vec{k} \cdot \vec{R}} = -\tilde{k} \sum_{\vec{R}' \text{ sosed.}} (e^{i\vec{k} \cdot \vec{R}} - e^{i\vec{k} \cdot \vec{R}'}) = -\tilde{k} e^{i\vec{k} \cdot \vec{R}} \sum_{\vec{R}' \text{ sosed.}} (1 - e^{i\vec{k} \cdot (\vec{R} - \vec{R}')})$$

$$\omega^2 M = \tilde{k} \sum_{\vec{R}' \text{ sosed.}} (1 - e^{i\vec{k} \cdot (\vec{R} - \vec{R}')})$$

$$\omega^2 = \frac{\tilde{k}}{M} \cdot \frac{1}{2} \left[\sum_{\vec{R}'} (1 - e^{i\vec{k} \cdot (\vec{R} - \vec{R}')}) + \sum_{\vec{R}'} (1 - e^{-i\vec{k} \cdot (\vec{R} - \vec{R}')}) \right] =$$

$$= \frac{\tilde{k}}{M} \sum_{\vec{R}'} (1 - \cos(\vec{k} \cdot (\vec{R} - \vec{R}'))) = \frac{2\tilde{k}}{M} \sum_{\vec{R}'} \sin^2 \frac{\vec{k} \cdot (\vec{R} - \vec{R}')}{2}$$

• ker imamo $\sin^2$, lahko vstavimo ta 3 sosedo in rezultat pomnožimo z 2

$$\omega^2 = \frac{4\tilde{K}}{M} \left[\sin^2\left(\frac{\vec{k} \cdot \vec{a}_1}{2}\right) + \sin^2\left(\frac{\vec{k} \cdot \vec{a}_2}{2}\right) + \sin^2\left(\frac{\vec{k} \cdot (\vec{a}_1 + \vec{a}_2)}{2}\right) \right]$$

$$|\vec{k}| \ll \frac{\pi}{a}$$

$$\begin{aligned} \omega^2 &= \frac{4\tilde{K}}{M} \left[\left(\frac{\vec{k} \cdot \vec{a}_1}{2} \right)^2 + \left(\frac{\vec{k} \cdot \vec{a}_2}{2} \right)^2 + \left(\frac{\vec{k} \cdot (\vec{a}_1 + \vec{a}_2)}{2} \right)^2 \right] = \frac{4\tilde{K}}{M} \left[(k_x a)^2 + \left(\frac{1}{2} k_x a + \frac{\sqrt{3}}{2} k_y a \right)^2 + \left(\frac{1}{2} k_x a - \frac{\sqrt{3}}{2} k_y a \right)^2 \right] \\ &= \frac{4\tilde{K}}{M} \left[(k_x a)^2 + \left(\frac{1}{2} k_x a + \frac{\sqrt{3}}{2} k_y a \right)^2 + \left(\frac{1}{2} k_x a - \frac{\sqrt{3}}{2} k_y a \right)^2 \right] = \frac{4\tilde{K}}{M} \left[(k_x a)^2 + \frac{1}{2} (k_x a)^2 + 2 \cdot \frac{3}{4} (k_y a)^2 \right] = \frac{3}{2} \frac{\tilde{K}}{M} k^2 a^2 \end{aligned}$$

$$\rightarrow \omega = \sqrt{\frac{3}{2} \frac{\tilde{K}}{M}} k a \propto k \Rightarrow \text{akustična}, \quad \omega = c k \rightarrow c = \sqrt{\frac{3}{2} \frac{\tilde{K}}{M}} a$$

$$H = \sum_{\vec{k}} \left(\frac{p_{\vec{k}}^2}{2M} + \frac{1}{2} M \omega^2(\vec{k}) x_{\vec{k}}^2 \right)$$

$$\rightarrow E = \sum_{\vec{k}} \left(n_{\vec{k}} + \frac{1}{2} \right) \hbar \omega(\vec{k}) \xrightarrow[\text{temperaturna razporeditev}]{\text{faktorijsko}} \sum_{\vec{k}} \left(e^{\beta \hbar \omega(\vec{k})} + \frac{1}{2} \right) \hbar \omega(\vec{k})$$

$$C(T) = \frac{dE}{dT}$$

$$E = \frac{S}{(2\pi)^2} \int_{ABC} \left(e^{\beta \hbar \omega(\vec{k})} + \frac{1}{2} \right) \hbar \omega d\vec{k}$$

• Visoka T: $\beta \hbar \omega \ll 1 \Rightarrow \beta \hbar \sqrt{\frac{K}{M}} \ll 1$

$$E = \frac{S}{(2\pi)^2} \int_{ABC} \frac{1}{\beta \hbar \omega(\vec{k})} \hbar \omega(\vec{k}) d\vec{k} = \left(\frac{1}{2} \text{ zanemarimo, ker bomo kasneje odvajali po T} \right)$$

$$= \frac{S}{(2\pi)^2} k_B T \int_{ABC} d\vec{k} = N k_B T \rightarrow C = N k_B$$

$\hookrightarrow$ št. atomov

* $\omega = ck$ velja le v sredini 1. BC, ampak je β tako velik, da bo eksponent odločil tisti del, kjer zveza $\omega = ck$ dolga več

• nizka T $k_B T \ll \hbar \sqrt{\frac{K}{M}} = \hbar \omega_0$

$$E = \frac{S}{(2\pi)^2} \int_{ABC} \frac{1}{e^{\beta \hbar \omega(\vec{k})} + 1} \hbar \omega(\vec{k}) d\vec{k} \approx \frac{S}{(2\pi)^2} \int_{ABC} \frac{1}{e^{\beta \hbar \omega(\vec{k})}} \hbar \omega(\vec{k}) d\vec{k} =$$

$$= \frac{S \hbar c}{(2\pi)^2} \int_0^{\omega_0} k^2 dk \frac{1}{e^{\beta \hbar c k} + 1} \cdot 2\pi = \frac{S \hbar c}{2\pi} \frac{1}{(\beta \hbar c)^3} \int_0^{\beta \hbar c \omega_0} u^2 \frac{1}{e^u + 1} du =$$

$$S_1 = a, \quad v_1 = a \frac{\sqrt{3}}{2} a = \frac{\sqrt{3}}{2} a^2$$

$$= \frac{S}{2\pi} \left(\frac{\hbar c}{k_B T} \right) \left(\frac{k_B T}{\hbar c} \right)^3 \cdot 2,4 = \frac{2,4 S}{3 \pi} \frac{M k_B^3}{a^3 \hbar^3} T^3 = \frac{2,4}{3 \pi} \frac{\sqrt{3}}{2} N \frac{M k_B^3}{\hbar^3} T^3$$

$$\rightarrow C = \frac{1,2 \sqrt{3}}{\pi} N \frac{M k_B^3}{\hbar^3} T^2 = N k_B \frac{1,2 \sqrt{3}}{\pi} \left(\frac{k_B T}{\hbar \omega_0} \right)^2$$

